

ORF307: Optimization

Precept: #8

Agenda

- Duality review
 - Motivation example
 - Weak/Strong duality
 - Farkas lemma
- Taking the dual example
- Dual of L1 norm

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Duality motivation & Obtaining lower bounds

We want to compute a lower bound (LB) of the following LP (**which is written in non-standard form**)

$$\begin{array}{ll}\min_x & x_1 + 3x_2 \\ s.t. & x_1 + x_2 \geq 2 \\ & x_2 \geq 1 \\ & x_1 - x_2 \geq 3\end{array}$$

Idea: add multiples of the constraints in such a way to **match the cost function**:

$$\left. \begin{array}{l} y_1(x_1 + x_2 \geq 2) \\ + y_2(x_2 \geq 1) \\ + y_3(x_1 - x_2 \geq 3) \end{array} \right\} \implies (y_1 + y_3)x_1 + (y_1 + y_2 - y_3)x_2 \geq 2y_1 + y_2 + 3y_3 \quad (1)$$

In order to match the cost function, the multipliers must satisfy the equality constraints:

$$y_1 + y_3 = 1 \quad \text{and} \quad y_1 + y_2 - y_3 = 3 \quad (2)$$

Also, to be able to multiply the constraints of the LP (which are $\geq$ inequalities) the multipliers must be nonnegative, i.e., we have the additional nonnegativity constraints:

$$y_1, y_2, y_3 \geq 0 \quad (3)$$

Putting all this together we can see that, from (1) we can derive the **dual objective**

$$2y_1 + y_2 + 3y_3 = b^T y$$

which serves as a **lower bound** to the the function

$$(y_1 + y_3)x_1 + (y_1 + y_2 - y_3)x_2.$$

In addition, the constraints (2) and (3) can serve as the constraints in the dual problem.

Now if you select the **feasible** dual vector $\hat{y} = (1, 2, 0)^T$ and substitute it to the dual objective in (1) we get $\hat{LB} = 4$ as lower bound on the original (primal) LP.

Similarly, if we use the **feasible** dual point $\tilde{y} = (0, 4, 1)^T$ we get $\tilde{LB} = 7$ as lower bound on the original (primal) LP.

NOTE: The larger lower bound ($\tilde{LB} = 7$) is better because it is closer to the optimal solution of the primal LP. In general, we are looking for those multipliers y_1^*, y_2^*, y_3^* that maximize the lower bound. This leads directly to the **dual LP**:

$$\begin{aligned} \max_y \quad & 2y_1 + y_2 + 3y_3 \\ \text{s.t.} \quad & y_1 + y_3 = 1 \quad (\text{Constraint for } x_1) \\ & y_1 + y_2 - y_3 = 3 \quad (\text{Constraint for } x_2) \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Observations:

- The right-hand side vector (rhs), $b = (2, 1, 3)^T$, of the constraints of the original (primal) LP have become the coefficients of the dual problem.
- The coefficients of the objective function of the primal problem have become the rhs vector of the constraints of the dual problem.
- The coefficient matrix of the primal LP problem

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$$

has been transposed to form the constraints of the dual problem.

- More formal and structured rules will be discussed in the following sections.

The optimal value of this dual problem provides the **tightest** lower bound possible, which, by the Strong Duality Theorem, is equal to the optimal value of the primal LP (if it is bounded).

Remark: Knowing or being able to compute a lower bound of a minimization LP problem is very useful. Imagine we are minimizing the cost of manufacturing some product or the minimizing the resource we need to use in the production process. Then it will be useful to know what is the least possible value the objective function can attain, given the constraints. But before we solve directly the original (i.e., the primal) minimization LP problem we could solve another LP (the dual LP), whose optimal solution provides a certified lower bound and is easier to compute.

Farkas' Lemma (aka, Theorem of Alternatives)

Farkas' Lemma states that for a given matrix $A \in \mathbb{R}^{m \times n}$ and vector $b \in \mathbb{R}^m$, exactly one of the following two statements is true:

- ⟨1⟩ $\exists x \in \mathbb{R}^n$ such that $Ax = b$ and $x \geq 0$ (feasibility of the primal problem).
- ⟨2⟩ $\exists y \in \mathbb{R}^m$ such that $A^T y \geq 0$ and $b^T y < 0$ (Certificate of Primal Infeasibility via dual unboundedness and hyperplane separation).

Proof Sketch

- **Statements ⟨1⟩ and ⟨2⟩ cannot both be True (that is they are mutually exclusive):**

We will show this by proof-by-contradiction. Assume on the contrary that both ⟨1⟩ and ⟨2⟩ are true. Then $\exists x \geq 0$ with $Ax = b$, and $\exists y$ with $A^T y \geq 0$. Multiplying $Ax = b$ by y^T from the left gives:

$$y^T b = y^T (Ax) = (y^T A)x$$

Since $A^T y \geq 0$ implies $y^T A \geq 0$, and we assumed $x \geq 0$, the product $(y^T A)x$ must be ≥ 0 , that is

$$y^T b = (y^T A)x \geq 0$$

But this contradicts the requirement in statement ⟨2⟩ that $b^T y < 0$. Therefore, ⟨1⟩ and ⟨2⟩ **cannot both be true**.

- **Statements ⟨1⟩ and ⟨2⟩ cannot both be False:**

This is proven using **Strong Duality**. Consider the (primal) LP problem:

$$\begin{array}{ll} \min & 0^T x \quad (\text{i.e., } 0) \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array}$$

Its dual LP problem is:

$$\begin{array}{ll} \max & -b^T y \\ \text{s.t.} & A^T y \geq 0 \end{array}$$

The key lies in analyzing the feasibility and boundedness of the primal (P) and dual (D) problems:

Case 1: Primal (P) is Feasible. Then:

- The system $Ax = b, x \geq 0$ is solvable (Statement $\langle 1 \rangle$ is True).
- Since the objective value is 0, $p^* = 0$ (P is feasible and bounded).
- By Strong Duality, the dual must also be feasible and bounded, with $d^* = p^* = 0$. Since $d^* = \max(-b^T y)$, an optimal value of 0 means that there is no feasible y (i.e., $A^T y \geq 0$) for which $-b^T y > 0$, or equivalently $b^T y < 0$. Thus, Statement $\langle 2 \rangle$ is False.

Case 2: Primal (P) is Infeasible.

- The system $Ax = b, x \geq 0$ is not solvable (that is, Statement $\langle 1 \rangle$ is False).
- The dual (D) is always feasible since $y = 0$ satisfies $A^T(0) = 0 \geq 0$. Thus, $d^* \neq -\infty$.
- If (P) is infeasible ($p^* = \infty$) and (D) is feasible, the only case possible from the Duality Theorem is that the dual must be **unbounded**, that is $d^* = \infty$.
- $d^* = \max(-b^T y) = \infty$ implies there exists a sequence of feasible y 's ($A^T y \geq 0$) such that $-b^T y \rightarrow \infty$. This means $b^T y \rightarrow -\infty$. Therefore, $\exists y$ such that $A^T y \geq 0$ and $b^T y < 0$. Thus, Statement $\langle 2 \rangle$ is True.

Since the primal problem must be either feasible or infeasible, exactly one of the two statements (1) or (2) must hold. ■

Geometric interpretation of Farkas lemma:

- **Case 1:** The vector b lies **in the cone generated by the columns of A** (the conical hull of A 's columns). This means that b can be expressed as a linear combination of the columns of the matrix A , that is $b = A_1 x_1 + A_2 x_2 + \dots + A_n x_n$ where the x_i 's are the coefficients and are non-negative.
- **Case 2:** The vector b lies outside the cone generated by the columns of A . This impossibility is certified by the existence of a vector y which defines a separating hyperplane $\{z | z^T y = 0\}$ that passes through the origin. Note that $A^T y \geq 0$ means that the entire cone lies on the non-negative side of the hyperplane thus proving it is separated from the cone. The vector b projects negatively onto the normal vector y .

Remark: The Farkas' lemma is fundamentally a **theorem of alternatives** for a system of linear constraints. It is one of the most important results in optimization, proving the ****non-existence**** of a primal solution by demonstrating the ****existence**** of a dual separating vector.

Remark: For convenience we also include the different cases between the primal and dual problems we discussed in the class, and are summarized in Table 1.

Dual (d^*)	Primal (p^*)		
	Infeasible ($p^* = \infty$)	Feasible & Bounded ($p^* < \infty$)	Unbounded ($p^* = -\infty$)
Unbounded ($d^* = \infty$)	Possible	Impossible	Impossible
Feasible & Bounded ($d^* < +\infty$)	Impossible	$p^* = d^*$ (strong duality)	Impossible
Infeasible ($d^* = -\infty$)	Exception	Impossible	Possible

Table 1: Different cases between primal and dual problems.

Dual of LP (standard and non-standard forms)

We know that for the primal LP problem (in standard form):

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

its dual LP problem is defined by:

$$\begin{aligned} \max_y \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c \geq 0 \end{aligned}$$

For the LP (which is **not** in standard form)

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned} \tag{4}$$

its dual problem is defined as

$$\begin{aligned} \max_y \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c = 0 \\ & y \geq 0 \end{aligned} \tag{5}$$

The dual problem (5) can be derived by considering the **Lagrangian function** of the primal problem (4)

$$L(x, y) = c^T x + y^T (Ax - b)$$

Note that, for any x that is primal feasible (i.e., x satisfies $Ax \leq b$), the constraint term $Ax - b$ is a vector of non-positive values (≤ 0). Because we have $y \geq 0$ and $(Ax - b) \leq 0$, their inner product is non-positive, i.e., $y^T (Ax - b) \leq 0$. Therefore, **for any feasible x** , we have:

$$L(x, y) = c^T x + y^T (Ax - b) \leq c^T x, \quad \forall x, y : Ax \leq b, y \geq 0$$

Remark: The Lagrangian acts as a function that always **sits below** the primal objective function, within the feasible region. You can think of $L(x, y)$ as a **lower bound generator**.

The **dual function** is defined as the minimum value of the Lagrangian over all $x \in \mathbb{R}^n$, not just the feasible x :

$$\begin{aligned} g(y) &= \min_{x \in \mathbb{R}^n} L(x, y) = \min_x (c^T x + y^T (Ax - b)) = \\ &= -b^T y + \min_x (c + A^T y)^T x \end{aligned}$$

- Note 1: the dual function is a function of y , because in the inner minimization problem we consider y as **fixed**. Hence we have $g : \mathbb{R}^m \rightarrow \mathbb{R}$.
- Note 2: because the minimization is performed over all $x \in \mathbb{R}^n$, the final result, $g(y)$, is simply a scalar value that represents **the best lower bound achieved by that particular choice of multiplier y** .

We can identify two cases:

- Case #1: if $c + A^T y \neq 0$, then $g(y) = -\infty$ (it is unbounded; this case is of no interest.)

This is easy to see, because when $c + A^T y \neq 0$ the function $(c + A^T y)^T x$ is linear with respect to $x \in \mathbb{R}^n$ and therefore

$$\min_{x \in \mathbb{R}^n} (c + A^T y)^T x = -\infty$$

(the values of every linear function with non-zero coefficient vector go from $-\infty$ to $+\infty$, and therefore it is unbounded from both directions). Then the dual function g becomes

$$g(y) = -b^T y - \infty = -\infty$$

- Case #2: if $c + A^T y = 0$, then we simply have $g(y) = -b^T y$. This case is the most interesting, because we now can define the dual function as

$$g(y) = -b^T y$$

We have shown that

$$g(y) = \begin{cases} -\infty, & \text{if } c + A^T y \neq 0 \\ -b^T y, & \text{if } c + A^T y = 0 \end{cases}$$

Then our task becomes to find the largest value (**maximize**) of the dual function, subject to the condition that the multipliers y are nonnegative. We can express the **dual problem** as the optimization problem:

$$\begin{aligned} \max_y \quad & g(y) \\ \text{s.t.} \quad & y \geq 0 \end{aligned}$$

Considering Case #2 (the interesting one) of the dual function, we can derive the **dual problem**:

$$\begin{aligned} \max_y \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c = 0 \\ & y \geq 0 \end{aligned}$$

Example:

Let's find the dual of the following LP:

$$\begin{aligned} \min_x \quad & 3x_1 + 4x_2 \\ \text{s.t.} \quad & x_1 + x_2 \geq 5 \\ & 2x_1 + x_2 \geq 6 \end{aligned}$$

This problem has $\geq$ inequalities in its constraints, so we multiply them with -1 to convert them to $\leq$ inequalities:

$$\begin{aligned} \min_x \quad & 3x_1 + 4x_2 \\ \text{s.t.} \quad & -x_1 - x_2 \leq -5 \\ & -2x_1 - x_2 \leq -6 \end{aligned}$$

Since we have transformed the original LP to the form described by (4) we can use the dual form described in (5):

$$\begin{aligned} \max_y \quad & 5y_1 + 6y_2 \\ \text{s.t.} \quad & -y_1 - 2y_2 + 3 = 0 \\ & -y_1 - y_2 + 4 = 0 \\ & y \geq 0 \end{aligned}$$

1 Dual of $L1$ norm problem

Consider the optimization problem:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & \|Ax + b\|_1 \leq 1 \end{aligned}$$

1. Formulate the above problem as an LP (with $\leq$ inequality constraints)

Solution:

The constraint can be equivalently written as

$$\|Ax + b\|_1 \leq 1 \iff \sum_{i=1}^m |a_i^T x + b_i| \leq 1$$

where a_i^T is the i -th row of the matrix A , and b_i the i -th element of b . By introducing the m auxiliary variables $s_i, i = 1, \dots, m$ (one for each $|a_i^T x + b_i|$) we get the LP

$$\begin{aligned} \min_{x,s} \quad & c^T x \\ \text{s.t.} \quad & s_i \geq a_i^T x + b_i, \quad i = 1, \dots, m \\ & s_i \geq -a_i^T x - b_i, \quad i = 1, \dots, m \\ & s_1 + \dots + s_m \leq 1 \end{aligned}$$

The above optimization is an LP, which can be written in a more compact form as follows:

$$\begin{aligned}
& \min_{x,s} \quad c^T x \\
& s.t. \quad Ax - s \leq -b \\
& \quad \quad -Ax - s \leq b \\
& \quad \quad \mathbf{1}^T s \leq 1
\end{aligned} \tag{6}$$

2. Derive the dual LP and show that it is equivalent to the optimization problem:

$$\begin{aligned}
& \max_z \quad b^T z - \|z\|_\infty \\
& s.t. \quad A^T z + c = 0
\end{aligned} \tag{7}$$

Solution:

Define $\tilde{x} = (x, s)^T$. Then the LP problem (6) can be written as

$$\begin{aligned}
& \min_{x,s} \quad \tilde{c}^T \tilde{x} \\
& s.t. \quad \tilde{A} \tilde{x} \leq \tilde{b}
\end{aligned}$$

where

$$\tilde{c} = \begin{pmatrix} c \\ 0 \end{pmatrix}, \quad \tilde{A} = \begin{bmatrix} A & -I \\ -A & -I \\ 0 & \mathbf{1}^T \end{bmatrix} \quad \text{and} \quad \tilde{b} = \begin{pmatrix} -b \\ b \\ 1 \end{pmatrix}$$

The dual problem then is defined as:

$$\begin{aligned}
& \max_{\tilde{y}} \quad -\tilde{b}^T \tilde{y} \\
& s.t. \quad \tilde{A}^T \tilde{y} + \tilde{c} = 0 \\
& \quad \quad \tilde{y} \geq 0
\end{aligned}$$

We can define the dual variable $\tilde{y}$ in terms of three other vectors as follows:

$$\tilde{y} = \begin{pmatrix} y \\ z \\ w \end{pmatrix}$$

If we expand the above dual LP we get the LP (in matrix-vector form):

$$\begin{aligned}
& \max_{y,z,w} \quad b^T y - b^T z - w \\
& s.t. \quad \begin{pmatrix} c \\ 0 \end{pmatrix} + \begin{bmatrix} A^T & -A^T & 0 \\ -I & -I & \mathbf{1} \end{bmatrix} \begin{pmatrix} y \\ z \\ w \end{pmatrix} = 0 \\
& \quad \quad y, z, w \geq 0
\end{aligned}$$

If we do some very simple algebra, the previous LP can be written as:

$$\begin{aligned}
& \max_{y,z,w} \quad b^T(y - z) - w \\
& s.t. \quad A^T(y - z) + c = 0 \\
& \quad \quad y + z = \mathbf{1}w \\
& \quad \quad y, z, w \geq 0
\end{aligned} \tag{8}$$

We note that for fixed y and z such that $A^T(y - z) + c = 0$ (i.e., y, z satisfy the constraint), the objective of the dual LP (8) is maximized, when the scalar variable w is minimized.

Now, let's define the vector $\tilde{z}$ such that

$$\tilde{z} = y - z \iff \tilde{z}_i = y_i - z_i \forall i$$

and combine it with the second equality constraint and the non-negativity constraints of the dual LP (8)

$$y_i - z_i = \tilde{z}_i \tag{9}$$

$$y_i + z_i = w \tag{10}$$

$$y_i, z_i \geq 0 \tag{11}$$

Adding (9) and (10) yields

$$2y_i = \tilde{z}_i + w, \quad \forall i \implies y_i = (w + \tilde{z}_i)/2, \quad \forall i$$

Similarly, subtracting (9) from (10) yields

$$2z_i = w - \tilde{z}_i \quad \forall i \implies z_i = (w - \tilde{z}_i)/2, \quad \forall i$$

Using (11) we can derive

$$y_i \geq 0 \implies (w + \tilde{z}_i)/2 \geq 0 \implies \tilde{z}_i \geq -w \quad \forall i$$

$$z_i \geq 0 \implies (w - \tilde{z}_i)/2 \geq 0 \implies \tilde{z}_i \leq w, \quad \forall i$$

Hence we have shown

$$-w \leq \tilde{z}_i \leq w, \quad \forall i \implies |\tilde{z}_i| \leq w, \forall i \implies \max_i \{|\tilde{z}_i|\} \leq w \implies \|\tilde{z}\|_\infty \leq w \tag{12}$$

From (12) we can see that the minimum (smallest) value that w can take is indeed $\|\tilde{z}\|_\infty$. We can then substitute w for $\|\tilde{z}\|_\infty$ in the dual problem (8) and obtain the simplified problem optimization problem (note that it is not an LP)

$$\begin{aligned} \max_z \quad & b^T \tilde{z} - \|\tilde{z}\|_\infty \\ \text{s.t.} \quad & A^T \tilde{z} + c = 0 \end{aligned} \tag{13}$$

Note that we have omitted the constraints $y + z = \mathbb{1}w$ and $y, z, w \geq 0$ because we used them to derive that $w = \|\tilde{z}\|_\infty$ and therefore they are implicitly included.

3. Give a direct argument that whenever x is primal feasible and z is dual feasible, we have

$$c^T x \geq b^T z - \|z\|_\infty$$

Solution:

We can give two arguments.

- (a) The first argument is to use weak duality. This is the **most commonly used and recommended** to use.
- (b) **(The second argument is more complicated but introduces a new inequality)**: The second argument uses the Holders inequality which generalizes the Cauchy-Schwartz inequality for any norm

$$\forall u, v \in \mathbb{R}^n, |u^T v| \leq \|u\|_p \|v\|_q$$

where $p, q > 0$ and $\frac{1}{p} + \frac{1}{q} = 1$. You can see that the Holder inequality includes the L_2 (Euclidean) norm, when $p = q = 2$. But we will use it for $p = 1$ and $q = \infty$. In this case we have $\frac{1}{1} + \frac{1}{\infty} = 1$ and then we get

$$\forall u, v \in \mathbb{R}^n, |u^T v| \leq \|u\|_1 \|v\|_\infty$$

Substituting $u = Ax + b$ and $v = z$ we get

$$(Ax + b)^T z \leq \|Ax + b\|_1 \|z\|_\infty \implies (Ax + b)^T z \leq \|z\|_\infty$$

because $\|Ax + b\|_1 \leq 1$. By rearranging the previous inequality we obtain

$$x^T A^T z + b^T z \leq \|z\|_\infty \implies b^T z - \|z\|_\infty \leq -x^T A^T z \quad (14)$$

Using the constraint of the optimization problem (13) we have

$$A^T z + c = 0 \implies x^T (A^T z + c) = 0 \implies c^T x = -x^T A^T z \quad (15)$$

Substituting (15) to (14) we obtain

$$b^T z - \|z\|_\infty \leq c^T x$$